1 Solution — GIAM 7.4 Problem 1

1.1 Problem

Use the binomial theorem (with $x = 1000$ and $y = 1$) to calculate 1001^6.

1.2 Setup — Binomial Theorem

The binomial theorem states

$$(x+y)^n \;=\; \sum_{k=0}^{n} \binom{n}{k} x^{n-k}\, y^k.$$

The hint sets $x = 1000$, $y = 1$, and $n = 6$, so

$$1001^6 \;=\; (1000+1)^6 \;=\; \sum_{k=0}^{6} \binom{6}{k} 1000^{6-k}\, 1^k \;=\; \sum_{k=0}^{6} \binom{6}{k} 1000^{6-k}.$$

Since $1^k = 1$ for every k, each term reduces to $\binom{6}{k} \cdot 1000^{6-k}$.

1.3 Step 1 — Read the Binomial Coefficients

Row 6 of Pascal's triangle gives the seven coefficients $\binom{6}{k}$:

$$\binom{6}{0} = 1, \quad \binom{6}{1} = 6, \quad \binom{6}{2} = 15, \quad \binom{6}{3} = 20, \quad \binom{6}{4} = 15, \quad \binom{6}{5} = 6, \quad \binom{6}{6} = 1.$$

(*Spot check:* the sum $1+6+15+20+15+6+1 = 64 = 2^6$, consistent with $\sum_k \binom{6}{k} = 2^6$ — i.e., $(1+1)^6 = 64$.)

1.4 Step 2 — Compute Each Term

The powers of 1000 are clean: $1000^m = 10^{3m}$.

k	$\binom{6}{k}$	1000^{6-k}	**Term** $\binom{6}{k} \cdot 1000^{6-k}$
0	1	10^{18}	1,000,000,000,000,000,000
1	6	10^{15}	6,000,000,000,000,000
2	15	10^{12}	15,000,000,000,000
3	20	10^{9}	20,000,000,000
4	15	10^{6}	15,000,000
5	6	10^{3}	6,000
6	1	1	1

Table 1.1

1.5 Step 3 — Add the Terms

Each term contributes a different range of digits, since the coefficients $\binom{6}{k}$ are all *less than* 1000 and each 1000^{6-k} shifts by exactly $\mathbf{3}$ decimal places. No carrying occurs across the boundaries. Column-stack the addition:

$$
\begin{array}{rr}
 & 1{,}000{,}000{,}000{,}000{,}000{,}000 \\
+ & 6{,}000{,}000{,}000{,}000{,}000 \\
+ & 15{,}000{,}000{,}000{,}000 \\
+ & 20{,}000{,}000{,}000 \\
+ & 15{,}000{,}000 \\
+ & 6{,}000 \\
+ & 1 \\
\hline
 & 1{,}006{,}015{,}020{,}015{,}006{,}001
\end{array}
$$

1.6 Conclusion

$$\boxed{1001^6 \;=\; 1{,}006{,}015{,}020{,}015{,}006{,}001}$$

1.7 Remark — Reading Pascal's Triangle Off the Decimal Expansion

Note the digit groups of the answer, separated by commas:

$$\underbrace{1}_{=\binom{6}{0}},\ \underbrace{006}_{=\binom{6}{1}},\ \underbrace{015}_{=\binom{6}{2}},\ \underbrace{020}_{=\binom{6}{3}},\ \underbrace{015}_{=\binom{6}{4}},\ \underbrace{006}_{=\binom{6}{5}},\ \underbrace{001}_{=\binom{6}{6}}.$$

Reading the digit blocks in order, we recover *Pascal's row* 6: $1, 6, 15, 20, 15, 6, 1$.

This is no coincidence. Multiplying $\binom{6}{k}$ by $1000^{6-k} = 10^{3(6-k)}$ shifts the coefficient $3(6-k)$ places to the left. Since each coefficient $\binom{6}{k}$ has at most 3 decimal digits (the maximum, 20, has 2), the shifted coefficients land in disjoint 3-digit slots — no carrying across slot boundaries. The decimal expansion of 1001^6 is therefore *literally* the coefficients of $(x+y)^6$ laid down at intervals of three digits.

1.8 Remark — When Does This Trick Stop Working?

The "Pascal-in-base-1000" trick works whenever every coefficient $\binom{n}{k}$ fits in 3 digits, i.e., $\binom{n}{k} < 1000$ for all k.

n	$\max_k \binom{n}{k}$	**Fits in 3 digits?**	**Pascal-pattern in 1001^n?**
6	20	yes	yes
9	126	yes	yes
12	924	yes (barely)	yes
13	1716	**no**	broken by carries
20	184,756	no	no

Table 1.2

So 1001^n reads off Pascal's row n for $n \leq 12$, and the visualization breaks at $n = 13$ where $\binom{13}{6} = 1716$ overflows the 3-digit block and carries into the next.

The same trick generalizes by changing the base. Working in base 10^d — i.e., using $x = 10^d$ — keeps the pattern alive as long as $\binom{n}{k} < 10^d$. For example, using $x = 10^6$ (i.e., computing $(10^6 + 1)^n$) extends the Pascal-readable range to $n \approx 23$ (since $\binom{23}{11} = 1{,}352{,}078 < 10^7$... actually that needs $d \geq 7$).

1.9 Remark — Why the Binomial Theorem is the Right Tool Here

A direct computation of 1001^6 by repeated multiplication requires manipulating 19-digit intermediate numbers. The binomial expansion, by contrast, decomposes the answer into 7 small, *non-overlapping* contributions — one per term, each easy to compute and trivial to assemble (because no carrying occurs).

The technique generalizes: any *near*-power-of-ten can be raised to small powers by binomial expansion, with the digits of the answer essentially "written down" rather than computed.

Quantity	**Useful split**	**Why it's clean**
1001^n	$(10^3 + 1)^n$	Pascal's row n at 3-digit intervals (for $n \leq 12$)
999^n	$(10^3 - 1)^n$	alternating signs of Pascal's row — sometimes overlap
101^n	$(10^2 + 1)^n$	Pascal's row at 2-digit intervals (for $n \leq 4$)
99^n	$(10^2 - 1)^n$	alternating-sign Pascal — needs care
10001^n	$(10^4 + 1)^n$	Pascal's row at 4-digit intervals (for $n \leq 16$)

Table 1.3

For this problem, 1001^6 is the *sweet spot*: 6 is small enough that the coefficients are small (≤ 20), and 1000 is a clean enough base that the shifts are by an integer number of digits.